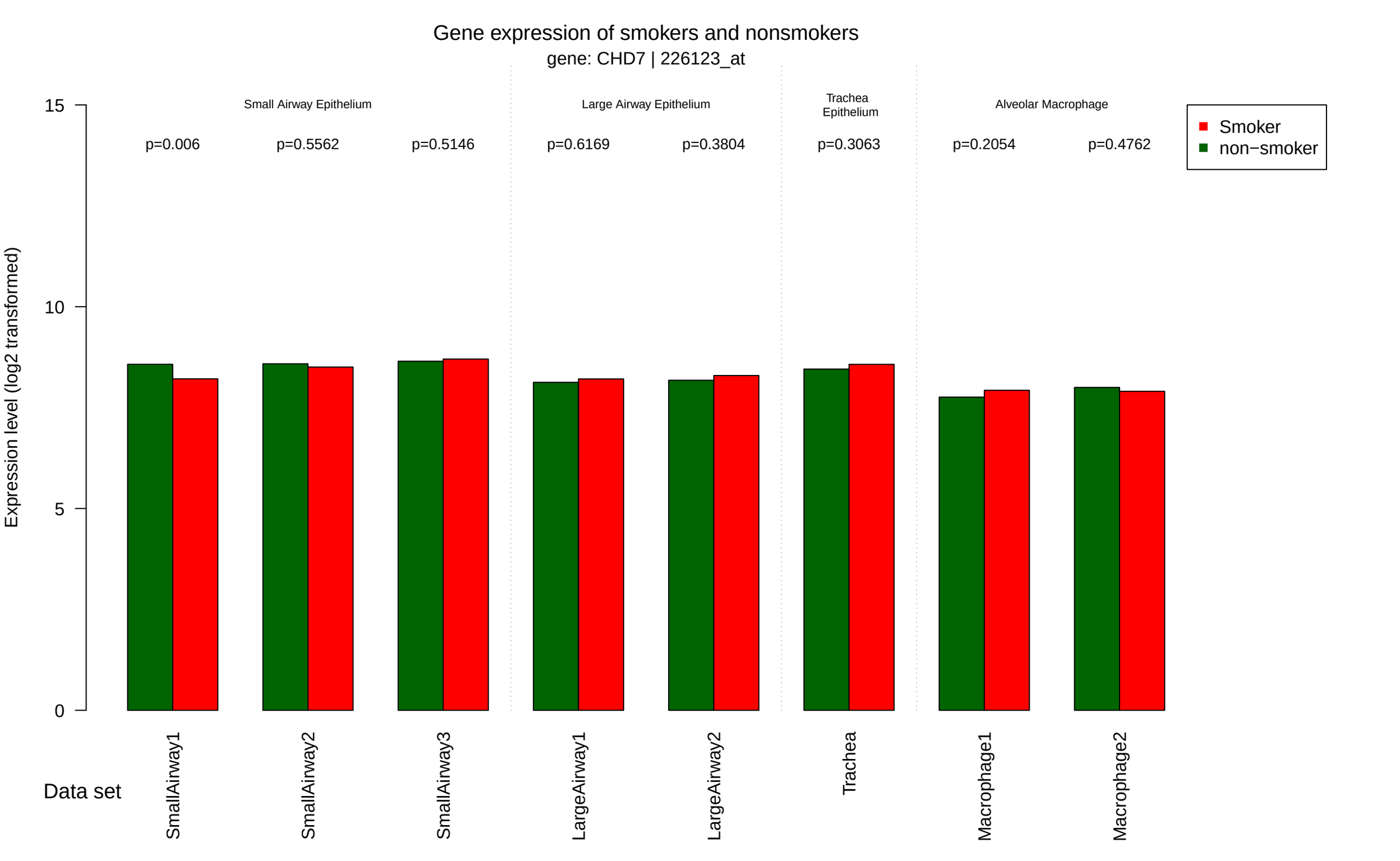
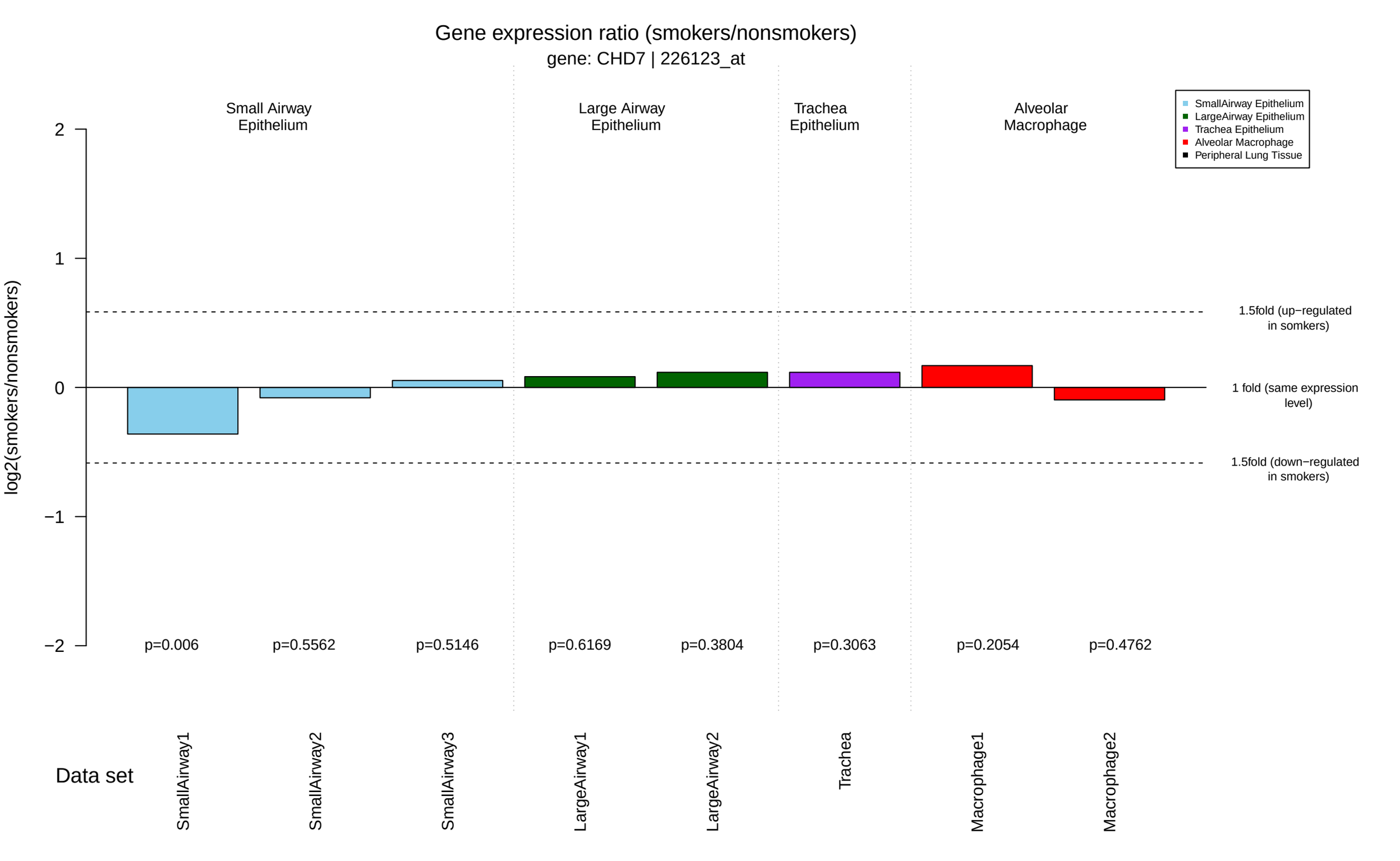
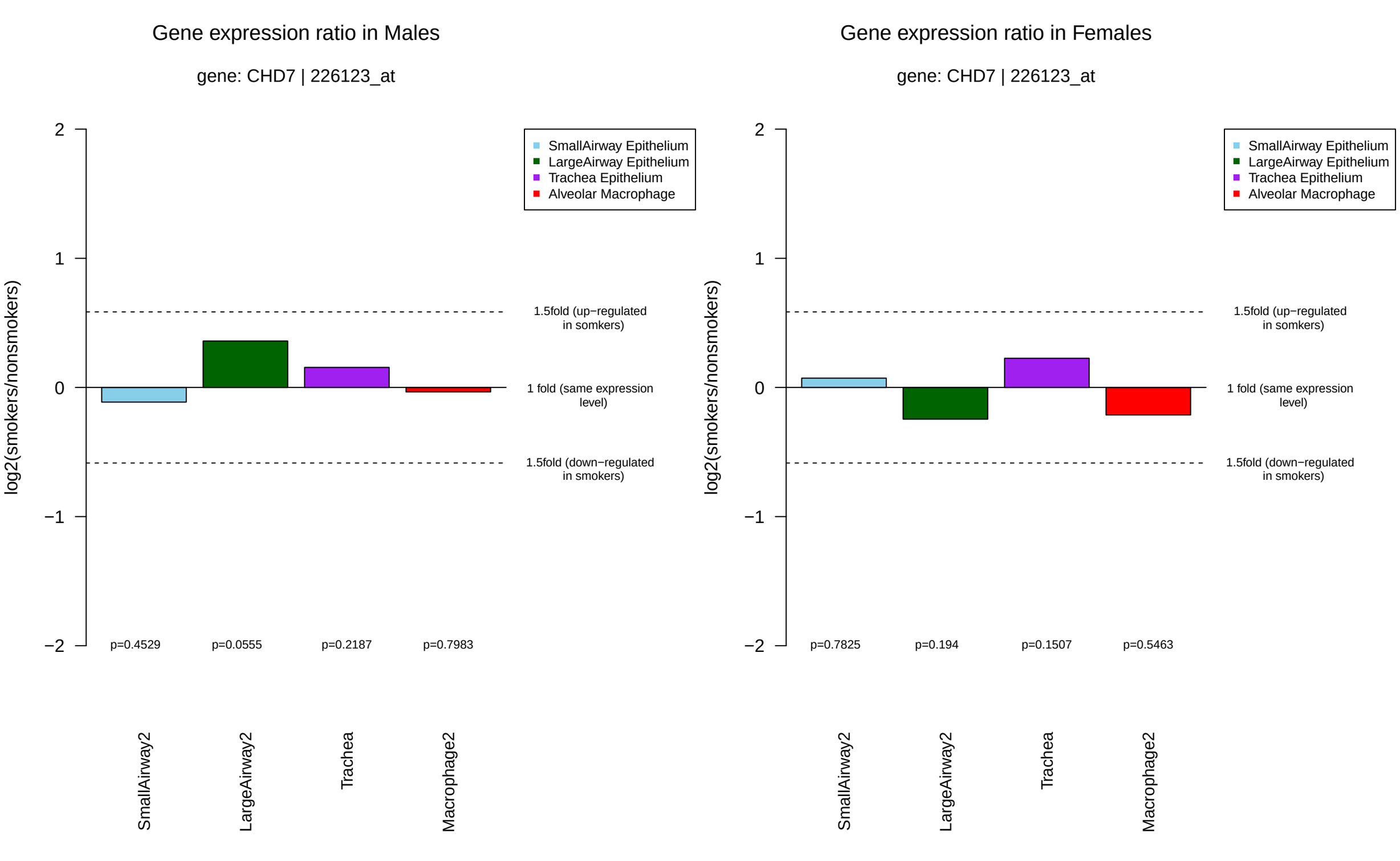
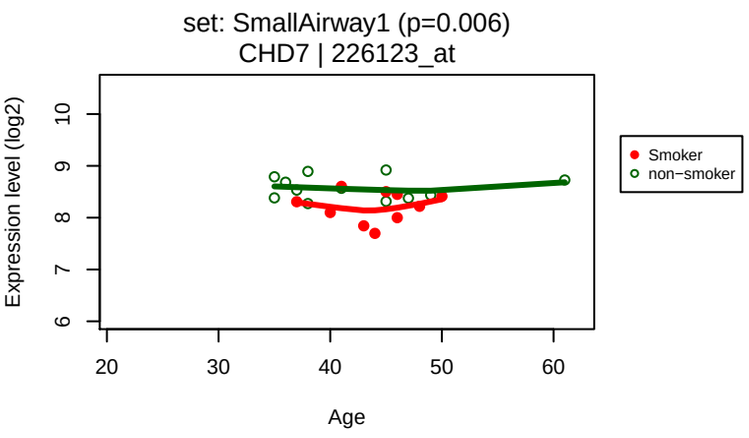
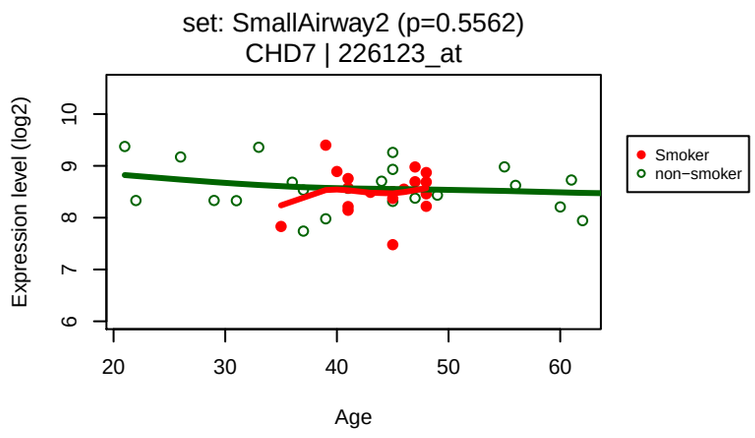
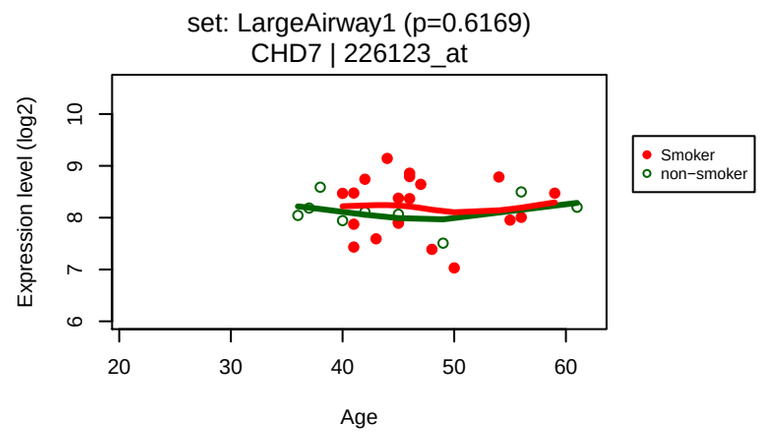

Gene expression of smokers and nonsmokers
gene: CHD7 | 226123_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.006
p=0.5562
p=0.5146
p=0.6169
p=0.3804
p=0.3063
p=0.2054
p=0.4762
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

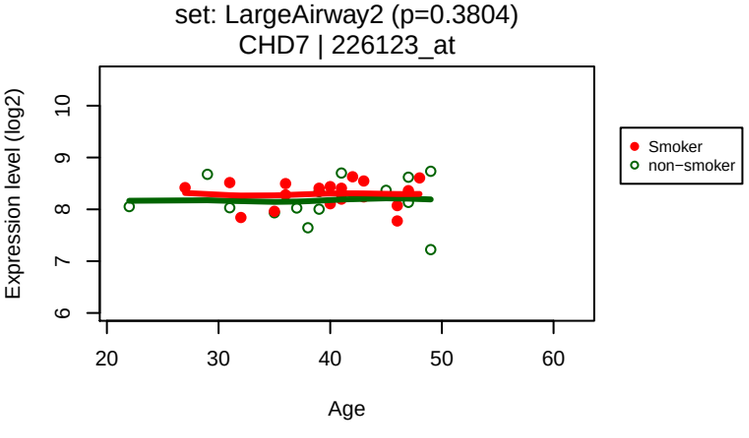

Gene expression ratio (smokers/nonsmokers)
gene: CHD7 | 226123_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.006
p=0.5562
p=0.5146
p=0.6169
p=0.3804
p=0.3063
p=0.2054
p=0.4762
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

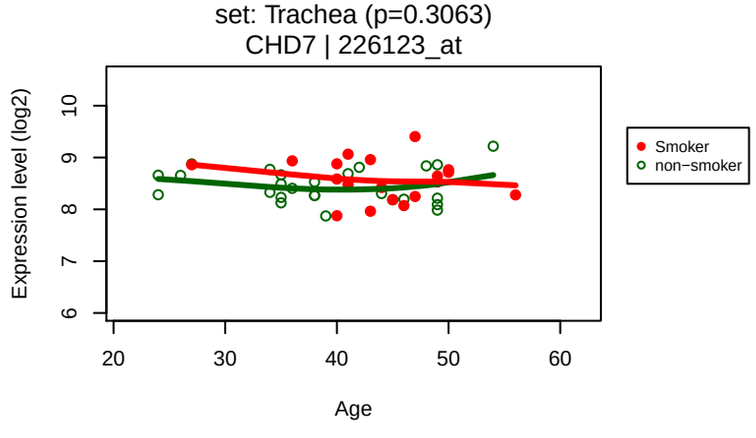

Gene expression ratio in Males
gene: CHD7 | 226123_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4529
p=0.0555
p=0.2187
p=0.7983
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CHD7 | 226123_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7825
p=0.194
p=0.1507
p=0.5463
SmallAirway2
LargeAirway2
Trachea
Macrophage2

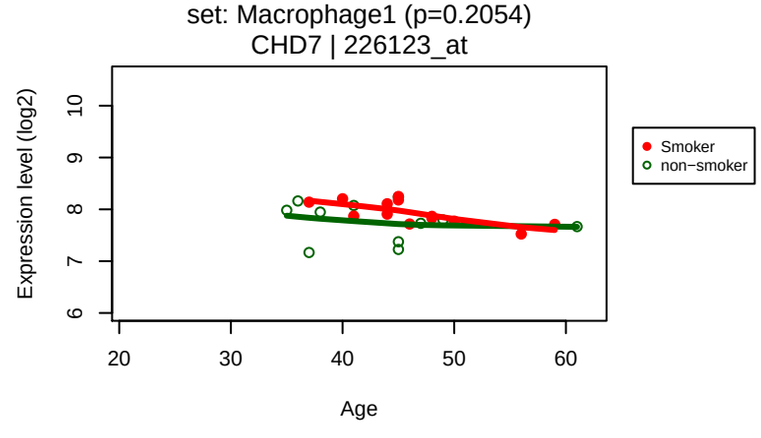

set: SmallAirway1 (p=0.006)
CHD7 | 226123_at
Expression level (log2)
Age
Smoker
non−smoker

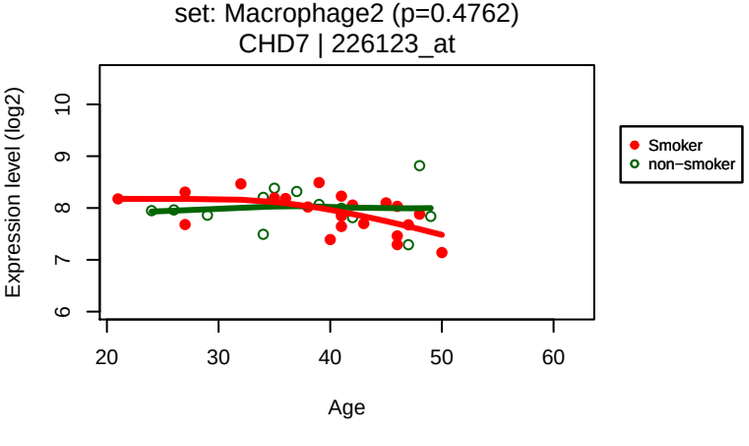

set: SmallAirway2 (p=0.5562)
CHD7 | 226123_at
Expression level (log2)
Age
Smoker
non−smoker

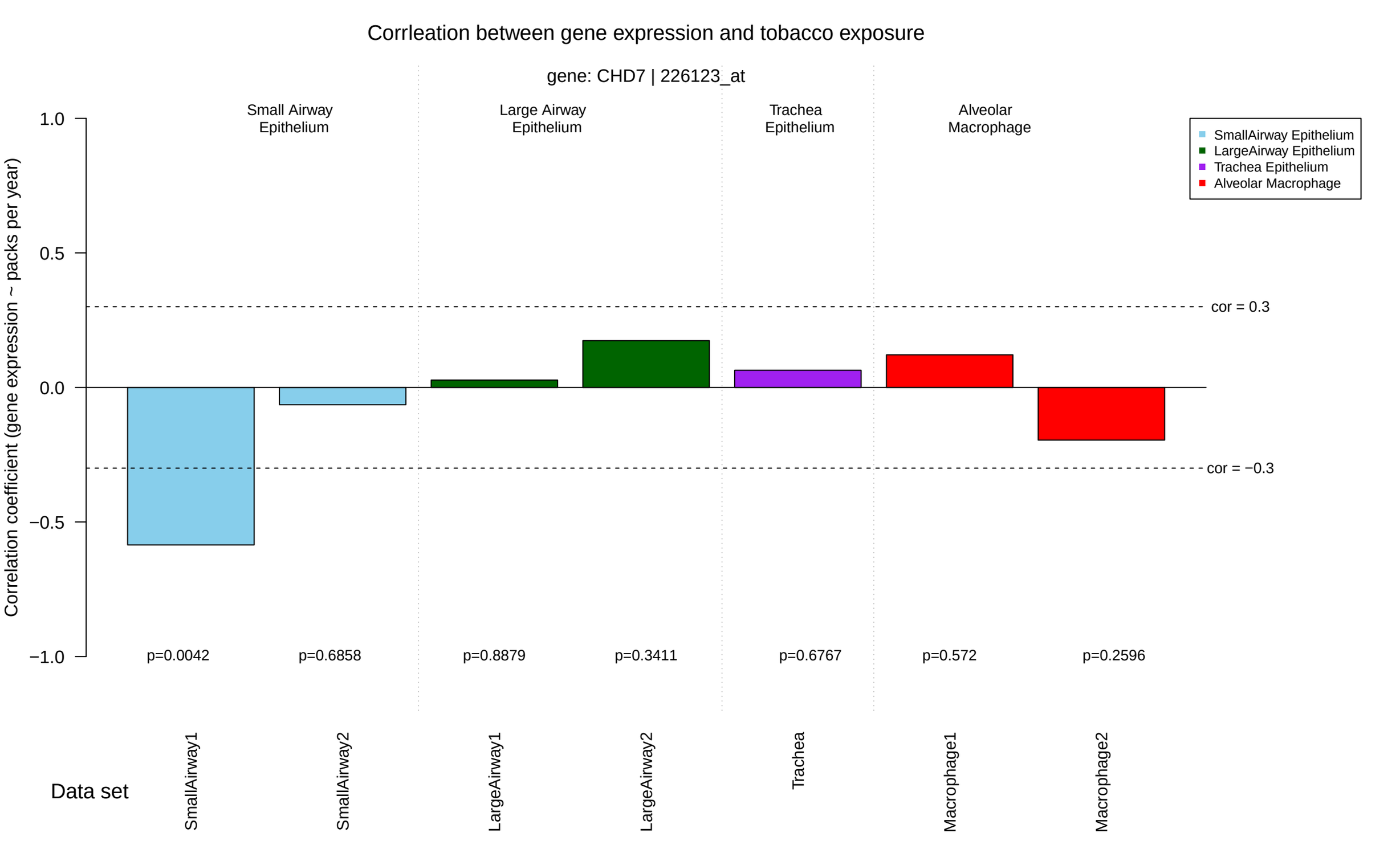

set: LargeAirway1 (p=0.6169)
CHD7 | 226123_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3804)
CHD7 | 226123_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3063)
CHD7 | 226123_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2054)
CHD7 | 226123_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4762)
CHD7 | 226123_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CHD7 | 226123_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0042
p=0.6858
p=0.8879
p=0.3411
p=0.6767
p=0.572
p=0.2596
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2